

accompany the paper.³² Furthermore, these confidence sets are quick to compute. Each sensitivity analysis plot in the empirical applications below took less than 9 minutes to compute on a 2012 Macbook Pro with a 2.3 GHz Quad-Core Intel Core i7 and 16 GB RAM.

6.1.3. Sensitivity analyses. Once the researcher has chosen a baseline class of restrictions on the possible violations of trends (e.g. relative magnitudes bounds $\Delta^{RM}(\bar{M})$ or smoothness bounds $\Delta^{SD}(M)$), we recommend conducting sensitivity analysis over the associated parameter ($\bar{M} \geq 0$ or $M \geq 0$, respectively) that governs how different the post-treatment violations of parallel trends can be from the pre-trends. It is natural to report both the sensitivity of the researcher's causal conclusion to the choice of this parameter and the "breakdown" parameter value at which particular hypotheses of interest can no longer be rejected; similar "breakdown" concepts appear in the partial identification settings of Horowitz and Manski (1995), Kline and Santos (2013), Manski and Pepper (2018), and Masten and Poirier (2020).³³ We illustrate how one can interpret the magnitudes of the breakdown points in our two empirical illustrations below.

6.2. Estimating the incidence of a value-added tax cut

Benzarti and Carloni (2019b, henceforth, BC) study the incidence of a decrease in the value-added tax (VAT) on restaurants in France. France reduced its VAT on sit-down restaurants from 19.6 % to 5.5 % in July of 2009. BC analyse the impact of this change using a dynamic DiD design that compares restaurants to a control group of other market services firms that were not affected by the VAT change, estimating the OLS regression specification

$$Y_{it} = \sum_{s \neq 2008} \beta_s \times 1[t = s] \times D_i + \phi_i + \lambda_t + \epsilon_{it}, \quad (19)$$

where Y_{it} is the log of (before-tax) profits for firm i in year t ; D_i is an indicator for whether firm i is a restaurant; ϕ_i and λ_t are firm and year fixed effects; and standard errors are clustered at the regional level. BC's main finding is that the VAT reduction had a large, positive effect on restaurant profits. Figure 4 shows the estimated event-study coefficients $\{\hat{\beta}_s\}$ from specification (19) (Benzarti and Carloni, 2019a). We can formally reject the hypothesis that $\beta_{pre} = 0$ ($p < 0.01$), as there appears to have been a difference in trends between 2006 and 2007. Nevertheless, the post-treatment coefficients for 2009–2011 appear to be substantially larger in magnitude than any of the pre-trends coefficients.

A key concern in this empirical setting is that there may be unobserved, industry-specific or macroeconomic shocks that would have affected restaurants differently from other market-services firms even in the absence of a change in VAT. It seems reasonable to impose that the industry-specific shocks to restaurants in the post-treatment period are not too much larger

32. The latest version of the R and Stata packages are respectively available at <https://github.com/asheshrambachan/HonestDiD> and <https://github.com/mcaceresb/stata-honestdid/>.

33. Our main focus in this paper is on constructing robust confidence sets given a particular restriction $\Delta(M)$, rather than inference on the identification breakdown point or breakdown frontier as in e.g. Masten and Poirier (2020). Note, however, that if we define $M^* = \min M$ s.t. $0 \in \mathcal{S}(\beta, \Delta(M))$ to be the identification breakdown point for a null effect, and $\hat{M}^* = \min M$ s.t. $0 \in \mathcal{C}(\hat{\beta}_n, \hat{\Sigma}_n; \Delta(M))$ to be the sample breakdown point, then $P(\hat{M}^* \geq M^*) \geq P(0 \in \mathcal{C}(\hat{\beta}_n, \hat{\Sigma}_n; \Delta(M^*)))$. It follows that $(-\infty, \hat{M}^*]$ is a valid $(1 - \alpha)$ -level confidence interval for M^* provided that our conditions for size control are satisfied for $\Delta(M^*)$. We suspect that our results could be extended to allow for uniform coverage of the breakdown frontier under additional regularity conditions, but leave this to future work.



FIGURE 4

Event-study coefficients $\{\beta_s\}$ for log profits, estimated using the event-study specification in (19).

than those in the pre-treatment period—whereas imposing that industry-specific shocks follow a smooth trend seems unreasonable—and so we base our analysis on bounds on relative magnitudes $\Delta^{RM}(\bar{M})$.

The left panel of Figure 5 shows robust confidence sets for the treatment effect in 2009 for $\Delta^{RM}(\bar{M})$ using different values of $\bar{M}$. The figure shows that if we impose $\bar{M} = 1$, meaning that we restrict the post-treatment violations of parallel trends to be no larger than the maximal pre-treatment violation of parallel trends, then we obtain a robust confidence set of $[0.07, 0.31]$ for the causal effect on restaurant profits in 2009. This is wider than the original OLS confidence interval which is only valid if parallel trends holds exactly, but nevertheless rules out a null effect on restaurant profits in 2009. Looking further to the right, we see that the “breakdown value” for a null effect is around $\bar{M} = 2$. Thus, our conclusion of a significant effect on restaurant profits depends on whether we are willing to restrict that the post-treatment violations of parallel trends can be no more than twice as large as the maximal pre-treatment violation. Given that the first year after the treatment coincided with a large recession in France (2009), it may be plausible that the differential factors affecting restaurants were larger in that year than in the pre-treatment period. Our approach helps formalize how much larger they would need to be to reject the conclusion of a null effect (or other hypotheses).

The right panel of Figure 5 shows analogous results when the estimand is the average causal effect on restaurant profits across all four post-treatment periods ($\bar{\tau}$). When $\bar{M} = 1$, our robust confidence set now includes zero, and is about twice as large as for the first-period effect. The intuition for why the confidence sets are larger when looking at $\bar{\tau}$ than τ_{2009} is that $\Delta^{RM}(\bar{M})$ bounds the violation of parallel trends across *consecutive periods* by $\bar{M}$ times the max in the pre-treatment period. Thus, the identified set will be larger for later periods, since the treatment and control groups have more time to diverge (e.g. the identified set for the second period will be twice as large as for the first period).³⁴ If we are willing to bound the magnitude of economic

34. This intuition holds generally for choices of Δ that bound changes across consecutive periods, but need not necessarily hold for other types of restrictions.

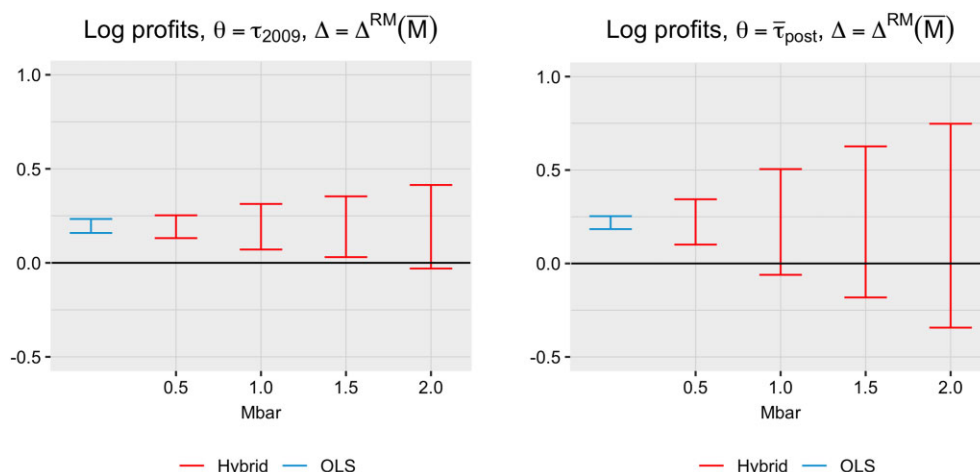


FIGURE 5
Sensitivity analysis for Benzarti and Carloni (2019b).

shocks by the max in the pre-treatment period, we will thus typically obtain wider confidence sets for parameters involving later periods.

Finally, in [Supplementary Material, Appendix C](#), we compare the confidence sets reported in Figure 5 to the sample analogue to the identified set, $\mathcal{S}(\hat{\beta}, \Delta)$, which we show is a Hausdorff consistent estimator of the identified set in this example. We find that the reported confidence sets are between 40 and 80% longer than the estimated identified set, suggesting that both sampling uncertainty and the length of the identified set play an important role in the width of the confidence sets.

6.3. The effect of duty-to-bargain laws on long-run student outcomes

[Lovenheim and Willen \(2019a\)](#), henceforth LW) study the impact of state-level public sector duty-to-bargain (DTB) laws, which mandated that school districts bargain in good faith with teachers' unions. LW examine the impacts of these laws on the adult labour market outcomes of people who were students around the time that these laws were passed, comparing individuals across different states and different birth cohorts to exploit the differential timing of the passage of DTB laws across states. The authors estimate the following OLS regression specification separately for men and women, using data from the American Community Survey (ACS),

$$Y_{sct} = \sum_{r=-11}^{21} D_{scr} \beta_r + X'_{sct} \gamma + \lambda_{ct} + \phi_s + \epsilon_{sct}. \quad (20)$$

Y_{sct} is an average outcome for the cohort of students born in state s in cohort c in ACS calendar year t . D_{scr} is an indicator for whether state s passed a DTB law r years before cohort c turned age 18.³⁵ The event-study coefficients $\{\hat{\beta}_r\}$ estimate the dynamic treatment effects (or placebo

35. $D_{sc,-11}$ is set to 1 if state s passed a law 11 years or more after cohort c turned 18. Likewise, $D_{sc,21}$ is set to 1 if state s passed a law 21 or more years before cohort c turned 18.

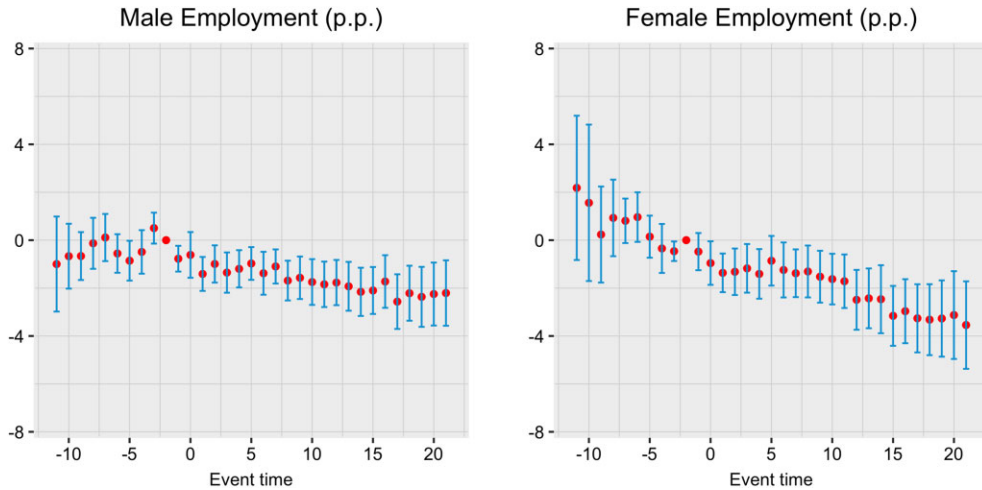


FIGURE 6

Event-study coefficient $\{\beta_r\}$ for employment, estimated using the event-study specification in (20).

effects) r years after DTB passage.³⁶ The remaining terms include time-varying controls, birth-cohort-by-ACS-year fixed effects, and state fixed effects. We normalize the event-study coefficient β_{-2} to 0.³⁷ We focus on the results where the outcome is employment.

Figure 6 plots the estimated event-study coefficients $\{\hat{\beta}_r\}$ from specification (20) (Lovenheim and Willen, 2019b). In the event-study for men (left panel), the pre-period coefficients are relatively close to zero, whereas the longer-run post-period coefficients are negative. By contrast, the results for women (right panel) suggest a downward-sloping pre-existing trend.

LW write that, the “primary concern in our identification strategy is the existence of secular trends that differ systematically with treatment” (p. 318), such as confounding changes in labour supply or educational attainment. Given that the concern is long-run trends that are likely to evolve smoothly over time, smoothness restrictions of the form Δ^{SD} seem natural in this context. Indeed, in some of their robustness checks, LW estimate models with group-specific linear trends, which roughly corresponds with the case $\Delta^{\text{SD}}(0)$.³⁸ It thus seems natural to consider relaxations of the form $\Delta^{\text{SD}}(M)$, which allows for deviations from non-linearity of no more than M between consecutive periods.

Figure 7 reports results for the treatment effect on employment for the cohort 15 years after the passage of a DTB law (as in Table 2 of LW), constructing robust confidence sets about how non-linear the difference in trends can be. In blue, we plot the original OLS confidence

36. Treatment timing in LW is staggered, and therefore the results in Sun and Abraham (2020) imply that β_r can be interpreted as a sensible weighted average of causal effects under parallel trends only if treatment effects are homogeneous across adoption cohorts. For simplicity, we focus on the robustness of the results to violations of parallel trends using the original specification in LW, which is valid under the assumption of homogeneous treatment effects. As discussed in Section 2.1, our sensitivity analysis can also be applied to estimators that are robust to treatment effect heterogeneity.

37. LW normalize event time -1 to 0, but discuss how cohorts at event time -1 may have been partially treated since LW impute the year that a student starts school with error. Since our robust confidence sets assume that there is no causal effect in the pre-period ($\tau_{pre} = 0$), we instead treat event-time -2 as the reference period in our analysis.

38. The two are not exactly equivalent, however, because LW include parametric trends into what they call a “parametric event-study” model (see their specification (2)), which imposes that treatment effects are linear in time since treatment, rather than the flexible dynamic event-study specification (20).